1-D Satellite Reaction Wheel

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Introduction

- **Reaction wheels** are used to maintain a system's orientation when an external torque is applied, via conservation of angular momentum.
  - Our model replicates a 1-D reaction wheel that is mounted on top of a satellite disk.
  - By using a motor to spin the reaction wheel, external torques applied to the satellite about the wheel's axis of rotation can be counteracted to maintain a stable attitude.

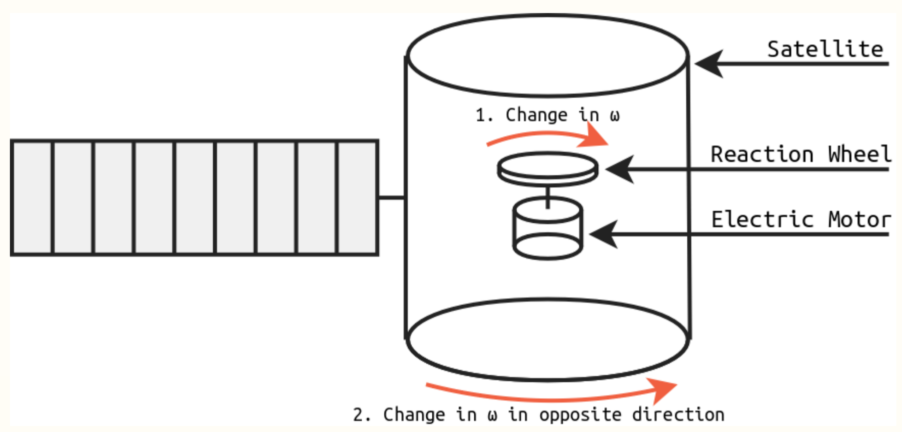

Figure 1. Reaction Wheel in a Satellite

Figure 2. Picture of Our Reaction Wheel

Motivation

- Reaction wheels are commonly used on satellites because they don't eat into the propellant Δv budget, extending the operational life of the satellite.
  - These wheels keep the satellite at a desired orientation and allow them to slew
- Two or more reaction wheels on different axes of rotation allow for multi-directional stability.
- It is important that reaction wheels controllers are finely tuned to smoothly slew satellites to a precise attitude and keep them there.

Hardware Setup

Components

- **Three Disks:** base, central satellite disk, and reaction wheel, all oriented on the same vertical axis
  - Reaction wheel has weights on the edges for greater angular momentum capacity, reducing the max motor RPM needed to control the system.
- **Controller:** Arduino Nano 33 IoT
  - Handles real-time commanding of the motor-driver based on readings from the IMU.
- **Gyroscopic Feedback:** BNO085 9-DoF IMU
  - Provides Arduino Nano with hardware-fused orientation data in the form of Euler angles (yaw, pitch, and roll).
- **Actuation System:** Stepper Motor & DRV8825
  - The stepper motor is driven by a DRV8825 current-limiting stepper motor driver.
- **Power:** 12V NiMH battery power motor and Arduino. IMU is powered via 3.3V Arduino logic.
- **Additional Components:**
  - Skateboard bearings: allow the satellite wheel to rotate freely relative to the fixed base.
  - Felt pads prevent the base from sliding on tables.

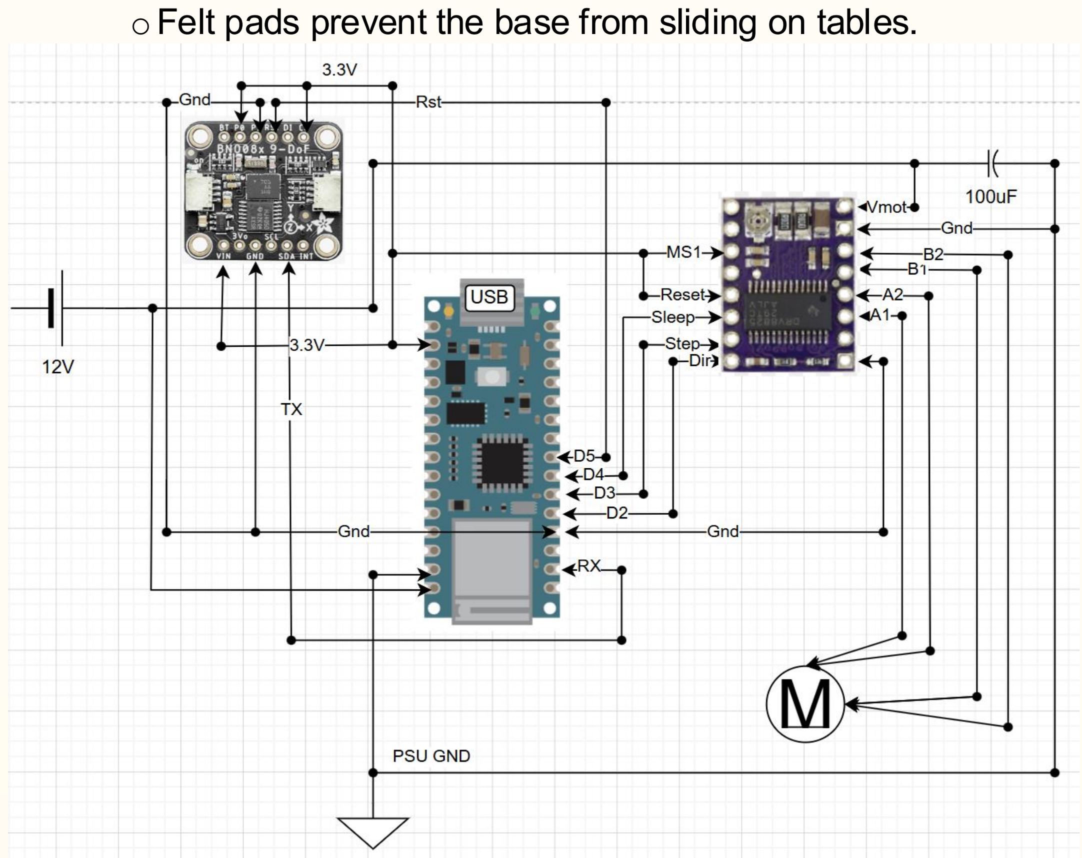

Figure 3. Circuit Diagram

System Modeling & Identification

- **Plant:** A 1-DoF satellite bus. The system is modeled as two coupled rotating masses: the reaction wheel and the satellite body, each with a theoretically-calculated rotational moment of inertia (MOI). The actuator is the stepper motor.
- **Controller:** Arduino Nano PID
- **Sensor:** IMU which provides feedback

Measured Variables & Actuators

- **Primary Sensor Input:** Yaw (θ)
  - The absolute orientation of the satellite relative to the start point. Used to calculate the angular rate.
- **Actuator Output:** Step Frequency (f)
  - Controls the speed of the reaction wheel, and angular momentum is a function of speed.

Controller Design

Controller Architecture

- **PID Controller**: A Proportional controller gain was included to reject disturbances to the desired displacement angle. (i.e., error between the desired angle and current angle). An integral controller was used to reduce the steady-state error. A derivative controller was included because we wanted to control overshoot and damping.

Stability

- To keep a stable closed-loop system, we first used our simulation to tune gains as a first pass. However, we expected the real system to vary from this significantly, so we achieved stability by tuning gains via trial & error with our real system.

Performance Criteria

- The goals were to not exceed 20% max overshoot with a <5-second settling time, specifically when subjected to step displacement angle disturbances between 10 and 30 degrees (these would be achieved by manual disturbance, so exact angles would vary and these impulses would never be a perfect step signal).

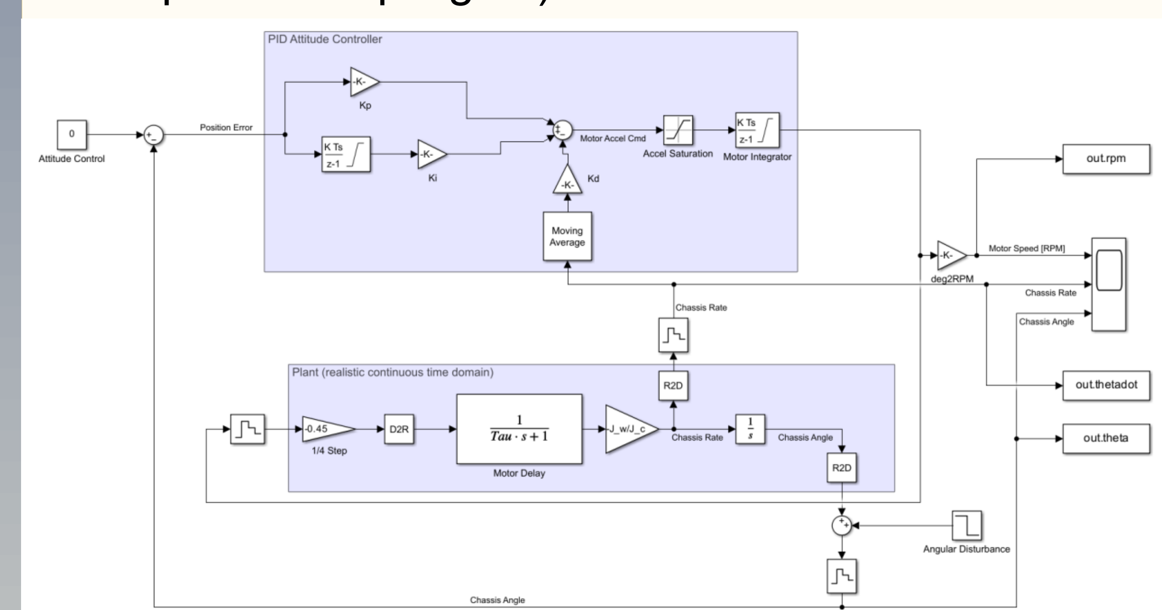

Figure 4. Simulink Diagram

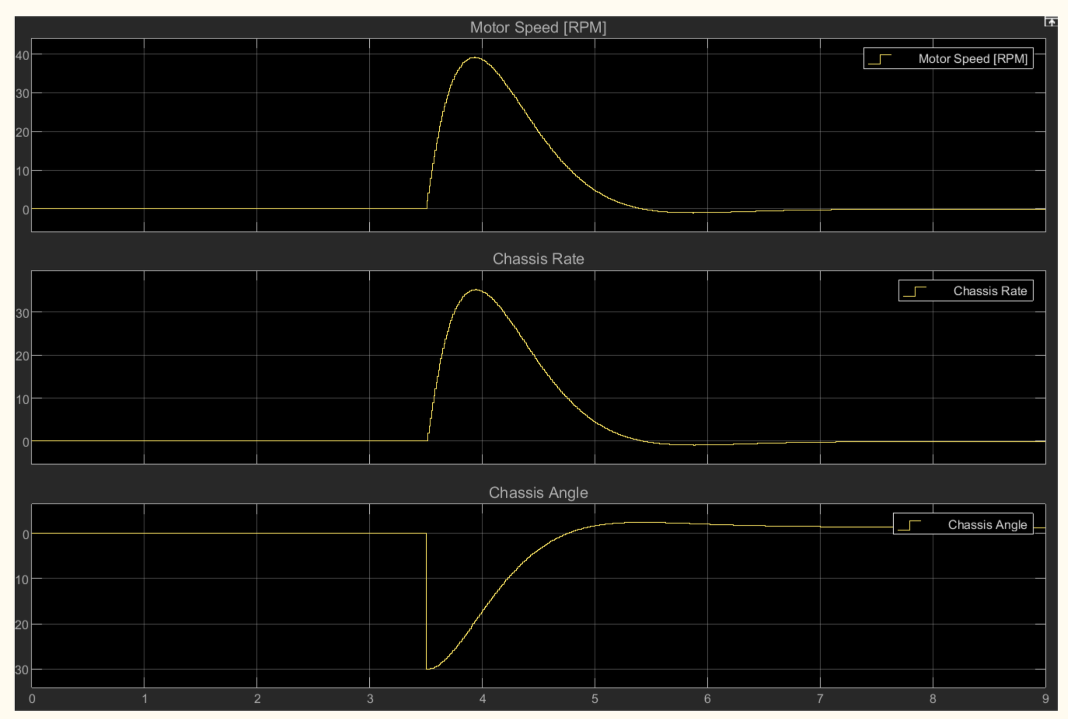

Figure 5. Simulink Model Response (30° Angular Step Input)

Evaluation and Results

- **Attitude Gains**
  - K_P = 40
  - K_I = 4
  - K_D = 24
- **Simulation Results**
  - Initial Disturbance 30.0°
  - Overshoot 2.41° (8.03%)
  - Settling Time 5.50 s
- **Measured (Actual) Results**
  - Initial Disturbance 31.59°
  - Overshoot 5.10° (16.14%)
  - Settling Time 2.76 s

Note: the large difference between the simulation and actual results are due to imperfect MOI theoretical calculations and the lack of bearing friction in our model.

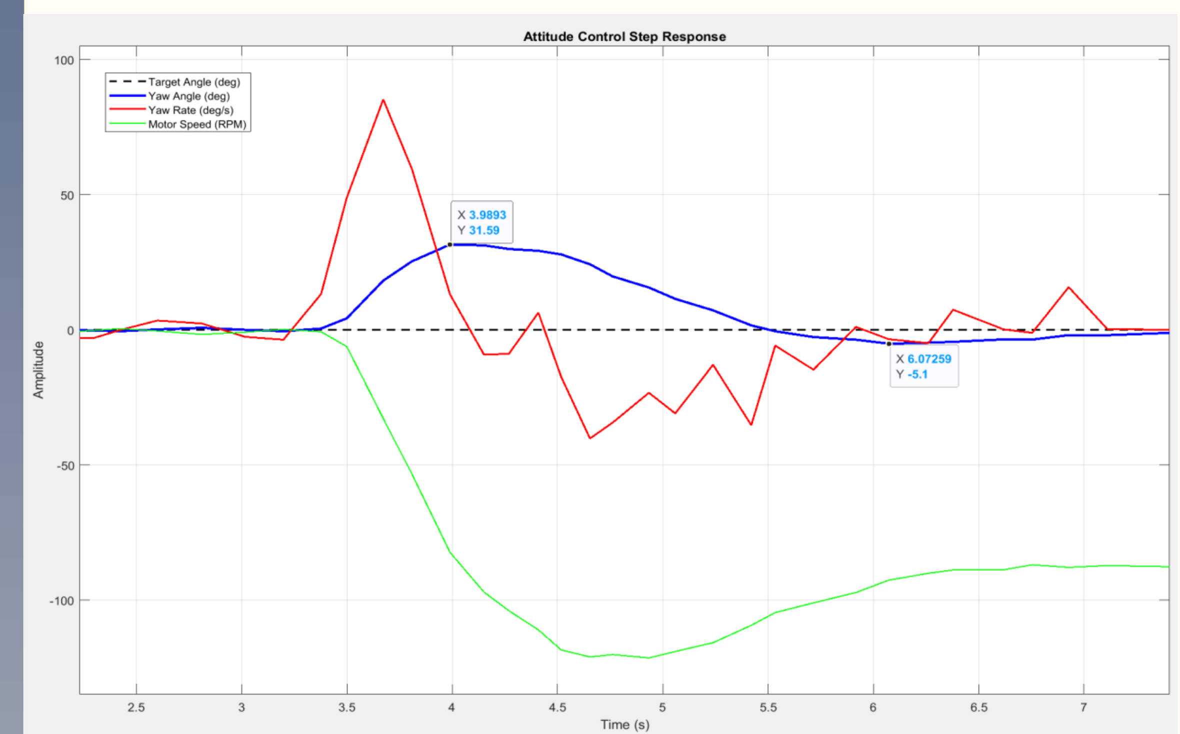

Figure 6. Empirical Satellite Response

Conclusion

- Prototype successfully demonstrated single-axis attitude control
- Results show the PID rejected disturbance torques with stability and efficiency
- More accurate moment of inertia values are required for an accurate simulation of the system
- Controller met performance requirements for overshoot and settling time
- Simulation was a bit outside the performance criteria set, but likely due to a slight variation in the simulated and actual design.

Future Efforts

- This reaction wheel only controls one axis of motion. A comprehensive reaction wheel subsystem would be comprised of 3+ wheels.
- Add additional complexity to the satellite's movement and controller setup.
- Additional tuning of gains could lead to reduced overshoot and decreased settling time, making the reaction wheel snappier.